Hamilton-Jacobi Theory as a Local Projection of Boundary-Kernel Composition

A non-invertibility theorem from global gluing sectors

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Abstract

This paper clarifies the relation between boundary-kernel composition and Hamilton-Jacobi theory. A composable boundary kernel assigns amplitudes or weights to bounded regions with prescribed boundary data and composes by integration over internal boundary data. In regular sharp or semiclassical regimes, such kernels can project to local Hamilton-Jacobi data: local action functions, endpoint response covectors, and the local Hamilton-Jacobi equation. The main result is that this projection is not invertible. On a configuration space with loops that cannot be shrunk to a point, one-component globally composable kernels can carry phase assignments to loop classes, written mathematically as characters $\chi : \pi_1(Q) \to U(1)$ from the fundamental group of Q to unit complex phases. Different phase assignments give inequivalent global kernels but the same local Hamilton-Jacobi data on patches with no such loops. The particle on a ring, equivalently the Aharonov-Bohm example, exhibits the mechanism: the local dynamics is free while the global kernel retains a holonomy parameter, a phase accumulated around a non-contractible loop. Thus Hamilton-Jacobi theory appears as a local projection of boundary-kernel composition, while global gluing sectors remain part of the full kernel. Together with the sharp-limit reconstruction of response geometry from composable region weights, this gives a precise sense in which boundary-kernel composition generalises Hamilton-Jacobi theory.

1 introduction

Hamilton-Jacobi theory turns a principal function into endpoint momenta and canonical response data [1]. Path-integral constructions, by contrast, assign amplitudes to histories and compose them by summing or integrating over intermediate data [2, 3]. On multiply connected configuration spaces those sums can carry global sectors labelled by the topology of paths, a standard point in the path-integral literature [4]. The Aharonov-Bohm effect is the familiar physical example of locally invisible holonomy with global consequences [5].

This paper uses that witness to ask whether a composable boundary kernel can be reconstructed from its local Hamilton-Jacobi data. The answer is no. The main result is a non-invertibility theorem for a projection: the map from composable kernels to local Hamilton-Jacobi action-response data forgets global gluing sectors.

The issue arises because composable region weights can, in a positive regular sharp limit, produce a sharp generator whose boundary derivatives define response covectors and a Lagrangian response relation [6]. That reconstruction overlaps with Hamilton-Jacobi geometry, and the overlap raises a natural inversion question: does the local action-response data determine the full kernel? The answer here is no. Hamilton-Jacobi geometry can appear as a local projection, but that projection does not determine the full composable kernel.

Combined with the sharp-limit reconstruction of response geometry from composable region weights [6], this non-invertibility result gives a precise sense in which boundary-kernel composition generalises Hamilton-Jacobi theory: Hamilton-Jacobi data arise as a local projection, while global gluing sectors remain part of the full kernel.

The paper studies scalar semiclassical kernels on a configuration space Q. A kernel is composable when the kernel for a union of regions is recovered by integrating the product of subregion kernels over the shared boundary data. For time intervals this is the familiar composition law

$$K_{T_2+T_1}(q_b, q_a) = \int_Q K_{T_2}(q_b, q) K_{T_1}(q, q_a)\ \mathrm{d}\mu(q).$$

In a local semiclassical regime the same kernel has local action functions S, more precisely local branches defined on a chosen patch, and their endpoint derivatives define response covectors. The local Hamilton-Jacobi projection Π_{HJ} retains this local action-response data. Standard Hamilton-Jacobi theory may of course be supplemented with global topological data if those data are supplied separately. The theorem concerns the local projection Π_{HJ} defined below: that projection alone does not contain the global sector.

The main theorem shows that Π_{HJ} is not injective. If Q is multiply connected, lifted paths carry global gluing classes. A scalar global sector is a character $\chi : \pi_1(Q) \to \mathrm{U}(1)$. Given a lifted local kernel on the universal cover, one obtains a family of kernels K_χ by summing over deck transformations with weight χ. Deck transformations label the different lifted homotopy classes of paths between the same projected endpoints. Multiplicativity of χ is exactly the condition that these weights compose under concatenation of global classes. Locally, on simply connected charts and short branches, the character is invisible to the Hamilton-Jacobi equation and to endpoint response covectors. Globally, different characters give different kernels.

Equivalently, the common local data define a nontrivial fiber of the projection:

$$\Pi_{\mathrm{HJ}}^{-1}(D_{\mathrm{loc}}) \supseteq \{K_\chi : \chi \in \mathrm{Hom}(\pi_1(Q), \mathrm{U}(1))\},$$

within the class of well-defined character-labelled kernels that share the same local branch actions. This fiber statement is the mathematical form of the paper's contribution: the inverse problem from local Hamilton-Jacobi data to a globally composable kernel is underdetermined.

The ring and Aharonov-Bohm example provide the concrete witness. The local Hamilton-Jacobi equation is the free-particle equation, while the global kernel and spectrum depend on a holonomy parameter. A flat global sector can be locally invisible and globally observable. The contribution here is to formulate that fact as a non-invertibility theorem for the projection from composable boundary kernels to local Hamilton-Jacobi data.

The structure is as follows. Section 2 defines the local Hamilton-Jacobi projection. Section 3 constructs the character-labelled kernel family and proves composition. Section 4 proves non-injectivity and states the boundary-completion corollary. Section 5 gives the ring/AB example. Section 6 records the scope of the result.

2 boundary kernels and the local Hamilton-Jacobi projection

This section fixes the kernel data, local branches, and projection used in the non-invertibility theorem.

Definition 1 (composable scalar boundary kernel)**.** *For time-interval regions on a configuration space Q, a scalar boundary kernel is a family $K_T(q_b, q_a)$, with $T > 0$ and $q_a, q_b \in Q$, such that gluing* *two intervals gives*

$$K_{T_2+T_1}(q_b, q_a) = \int_Q K_{T_2}(q_b, q) K_{T_1}(q, q_a)\ \mathrm{d}\mu(q)$$

whenever the integral is defined. The point q is the internal boundary datum exposed by the cut.

This time-interval definition is the simplest instance of the region-weight composition used in the sharp-limit reconstruction of Ref. [6]. A more general region formulation would replace the intermediate point q by data on an internal boundary. The theorem below uses the time-interval case.

Definition 2 (local semiclassical branch)**.** *Let $U \subset Q$ be simply connected. A kernel has a local semiclassical branch on U when, for endpoints in U and in the chosen time range, it has a leading oscillatory form*

$$K_T^\hbar(q_b, q_a) \sim a_T(q_b, q_a; \hbar) \exp\left(\frac{i}{\hbar} S_T(q_b, q_a)\right),$$

where S_T is a smooth real local action branch. The associated endpoint response covectors are

$$p_a = -\,\mathrm{d}_{q_a} S_T(q_b, q_a), \qquad p_b = \,\mathrm{d}_{q_b} S_T(q_b, q_a).$$

Definition 3 (local Hamilton-Jacobi projection)**.** *The local Hamilton-Jacobi projection $\Pi_{\mathrm{HJ}}(K)$ is the collection, over simply connected coordinate charts and admissible short-time branches, of:*

1. *the local action branches S_T, up to locally constant phase choices;*
2. *the endpoint response covectors (p_a, p_b);*
3. *the local Hamilton-Jacobi equation satisfied by S_T, when such an equation is available.*

Equivalently, it is the projection

$$\Pi_{\mathrm{HJ}} : K \longmapsto D_{\mathrm{loc}}(K) := \{S_{\mathrm{loc}},\ \mathrm{d}S_{\mathrm{loc}},\ \mathrm{HJ}_{\mathrm{loc}}\},$$

where D_{loc} denotes the collection of local action branches, their endpoint differentials, and the local Hamilton-Jacobi equations on the chosen charts and branches. The projection intentionally forgets global homotopy labels, holonomy characters, flat line-bundle data, Maslov data, fluctuation prefactors, and any record-probability rule beyond the local leading action-response data.

The word "projection" is literal. It is a map from a richer kernel-level object to a smaller local differential object. The next sections show that it has no inverse even in elementary scalar examples.

3 character-labelled kernels on a multiply connected space

Let Q be a connected smooth configuration space with universal cover $\pi : \widetilde{Q} \to Q$. Let Γ be the deck transformation group, identified with $\pi_1(Q)$ after choosing a base point. The convention below uses a left action of Γ on $\widetilde{Q}$. Write χ for a unitary character,

$$\chi : \Gamma \to \mathrm{U}(1), \qquad \chi(\gamma_2\gamma_1) = \chi(\gamma_2)\chi(\gamma_1).$$

Assumption 1 (lifted kernel and measure convention)**.** *Assume that $\widetilde{K}_T^\hbar(\tilde{q}_b, \tilde{q}_a)$ is a scalar lifted kernel on $\widetilde{Q}$ with the following properties.*

1. *It composes on the cover with respect to a measure $\mathrm{d}\widetilde{\mu}$.*

2. *The measure projects to* $\mathrm{d}\mu$ *on* Q*, and integration over* Q *can be represented as integration over a fundamental domain in* $\widetilde{Q}$*.*

3. *The lifted kernel is invariant under the diagonal deck action:* $\widetilde{K}^{\hbar}_{T}(\delta\tilde{q}_b, \delta\tilde{q}_a) = \widetilde{K}^{\hbar}_{T}(\tilde{q}_b, \tilde{q}_a)$*.*

4. *It has local semiclassical branches.*

5. *The image sums below converge, or are understood in the same controlled distributional or semiclassical sense in which the lifted kernel is used.*

Definition 4 (the K_χ family)**.** *For* $q_a, q_b \in Q$*, choose lifts* $\tilde{q}_a, \tilde{q}_b \in \widetilde{Q}$*. Define*

$$K^{\hbar}_{\chi,T}(q_b, q_a) = \sum_{\gamma\in\Gamma} \chi(\gamma)\, \widetilde{K}^{\hbar}_{T}(\tilde{q}_b, \gamma\tilde{q}_a).$$

Throughout the theorem the endpoint lifts are fixed once and for all. Changing the lift convention conjugates the class labelling and multiplies endpoint representatives by the corresponding phase convention; it does not change the non-injectivity statement.

The formula is the standard image-sum form of a kernel on a multiply connected space, with a character weighting the global class. In the present argument, the character is kernel data that is lost when only local Hamilton-Jacobi data are retained.

Lemma 1 (composition forces multiplicative class weights)**.** *Under Assumption 1, the family* $K^{\hbar}_{\chi,T}$ *composes under interval gluing whenever* χ *is a character. Conversely, if nonzero normalized scalar class weights* $c(\gamma)$*, with* $c(e) = 1$*, give a globally composable image-sum kernel for all concatenations of classes, then*

$$c(\gamma_2\gamma_1) = c(\gamma_2)c(\gamma_1).$$

Proof. In the composed kernel one sums over an intermediate point in Q. By Assumption 1, this integral can be lifted to a fundamental domain in $\widetilde{Q}$. The two image-sums then become a double sum over classes γ_1, γ_2. The lifted composition law combines the two lifted kernels into a lifted kernel whose global class is $\gamma_2\gamma_1$. The coefficient of that class is $\chi(\gamma_2)\chi(\gamma_1)$, which equals $\chi(\gamma_2\gamma_1)$ because χ is a character. Reindexing the double sum by the composite class gives the single image-sum for $K^{\hbar}_{\chi,T_2+T_1}$.

For the converse, exact composition for arbitrary concatenations requires the coefficient assigned to a composite class to equal the product of the coefficients assigned to the two pieces. With nonzero normalized coefficients and $c(e) = 1$, this is precisely multiplicativity. □

4 non-invertibility of the Hamilton-Jacobi projection

Lemma 2 (local invisibility of characters)**.** *Let* $U \subset Q$ *be simply connected. Restrict to endpoints and time ranges for which the local branch under consideration lifts to a path contained in one chosen lift of* U*. Then* $\Pi_{\mathrm{HJ}}(K_\chi)$ *on that branch is independent of* χ*.*

Proof. On the chosen branch the lifted path carries the identity global class. The character of the identity is one. Hence the leading local action branch and its endpoint derivatives are those of the lifted local kernel and do not depend on χ. If a local Hamilton-Jacobi equation is satisfied by the lifted action branch, the same equation is satisfied for every χ. □

Theorem 1 (the local Hamilton-Jacobi projection is not injective)**.** *Suppose Q has two distinct unitary characters of $\pi_1(Q)$, and suppose the corresponding image-sum kernels are well defined under Assumption 1. If two characters $\chi \neq \chi'$ give distinct kernels, then*

$$\Pi_{\mathrm{HJ}}(K_\chi) = \Pi_{\mathrm{HJ}}(K_{\chi'}) \quad \textit{locally,}$$

but $K_\chi \neq K_{\chi'}$ globally. Therefore the projection from globally composable scalar kernels of this form to local Hamilton-Jacobi data is not injective. If, in addition, the declared observable class distinguishes a relative global class on which χ and χ' differ, then the non-injectivity is physically visible.

Equivalently, if D_{loc} denotes the common local data, then

$$\Pi_{\mathrm{HJ}}^{-1}(D_{\mathrm{loc}}) \supseteq \{K_\chi : \chi \in \mathrm{Hom}(\pi_1(Q), \mathrm{U}(1))\},$$

within the class of well-defined character-labelled kernels sharing those local branches. The fiber is nontrivial whenever at least two such characters give distinct global kernels.

Proof. By Lemma 2, every K_χ in the family has the same local Hamilton-Jacobi projection on simply connected charts and admissible short branches. By Lemma 1, each character-labelled family is globally composable. If $\chi \neq \chi'$ and the resulting kernels are distinct, the global kernel contains data not present in Π_{HJ}. Hence Π_{HJ} is not injective on this class. If observables can distinguish interference between a class on which the characters differ and another class, the difference is not merely formal but physically visible. □

Definition 5 (boundary completion for this paper)**.** *For a declared class of kernels, cuts, and observables, a proposed boundary datum is complete when gluing over that datum reconstructs the whole kernel for every allowed cut in the class. If the gluing law fails after some datum has been projected away, a boundary completion is additional boundary data that restores the gluing law, modulo variables that are null or pure gauge for the declared observables.*

Corollary 1 (boundary-completion interpretation)**.** *For the scalar topological class above, local Hamilton-Jacobi data are an incomplete boundary proposal for global kernel composition. The missing character χ is boundary-completion data: it is additional data needed to reconstruct the globally composable kernel from local action-response data. If no observable in the declared class distinguishes χ, then χ is a null or gauge redundancy for that declared class and should be quotiented out rather than counted as physical boundary content.*

In this scalar reversible amplitude setting, the data erased by the local Hamilton-Jacobi projection are exactly the global character data needed for composition of path classes. Other settings can require different completion data; the theorem here identifies the scalar topological case.

5 global gluing classes

The preceding construction can be stated without committing to the ring example. Suppose the local Hamilton-Jacobi projection determines simply connected branch actions but the full histories carry global gluing classes. Write G_{glue} for the group of such classes when there is one base object, and for the corresponding gluing-class groupoid when boundary branch data vary. Composition in G_{glue} is concatenation of history fragments across cuts.

In the scalar case, a completed kernel assigns a phase weight $c(g)$ to a global class $g \in G_{\text{glue}}$. Exact gluing requires the class weight of a composite history to equal the product of the weights of its pieces:

$$c(g_2 g_1) = c(g_2)c(g_1).$$

Thus scalar global completion data are characters of the gluing-class group.

Theorem schema 1 (global gluing-class completion)**.** *Assume local Hamilton-Jacobi data determine only simply connected branch actions, while global kernel composition decomposes histories into classes in G_{glue}. If a scalar amplitude kernel is completed by assigning nonzero normalized weights to these classes, and if exact cut composition is required for arbitrary concatenations, then the weights form a character*

$$c : G_{\text{glue}} \to \mathrm{U}(1).$$

For a gluing-class groupoid or an internal non-scalar sector, the corresponding completion is a unitary representation of the relevant path or gluing-class groupoid, whenever the system is closed, reversible, and amplitude-valued.

This schema records the abstract mechanism behind the scalar theorem: local sharp dynamics can leave global composition underdetermined, and exact gluing then selects representation data for the classes that were erased by the local projection. The ring case below is the specialization $G_{\text{glue}} = \pi_1(S^1) = \mathbb{Z}$.

6 ring and Aharonov-Bohm example

Let $Q = S^1$, with angular coordinate θ, radius R, and universal cover $\mathbb{R}$. The deck group is $\Gamma = \mathbb{Z}$. Its scalar characters are

$$\chi_\alpha(n) = e^{2\pi i \alpha n}, \qquad \alpha \in \mathbb{R}/\mathbb{Z}.$$

The corresponding free-particle ring kernel can be written

$$K^{\hbar}_{\alpha,T}(\theta_b, \theta_a) = \left(\frac{mR^2}{2\pi i \hbar T}\right)^{1/2} \sum_{n\in\mathbb{Z}} \exp\left[\frac{i}{\hbar}\frac{mR^2(\theta_b - \theta_a + 2\pi n)^2}{2T}\right] e^{2\pi i \alpha n}.$$

The integer n labels winding. Locally, for short arcs, the Hamilton-Jacobi equation is the free-particle equation

$$\partial_T S + \frac{1}{2mR^2}(\partial_\theta S)^2 = 0.$$

The parameter α is not determined by that local equation.

Nevertheless the global kernel depends on α. In the usual Aharonov-Bohm interpretation, α is the dimensionless flux modulo an integer, up to sign convention. The sign depends on the orientation chosen for winding and on the convention used for the flux coupling. The energy spectrum has the familiar form

$$E_k(\alpha) = \frac{\hbar^2}{2mR^2}(k + \alpha)^2, \qquad k \in \mathbb{Z},$$

with the same convention dependence for the sign of α. The local Hamilton-Jacobi dynamics is free, while the global amplitude retains holonomy data.

This example realizes Theorem 1 in the smallest standard setting: local Hamilton-Jacobi data do not determine the full boundary kernel.

7 consequences beyond the ring

The ring example is the simplest scalar witness, but the non-invertibility mechanism is not tied to that geometry. The same projection issue appears whenever the leading local action and endpoint response data forget global or algebraic sectors needed for composition. Flat holonomy sectors and twisted boundary conditions give immediate variants of the scalar character story: local differential dynamics can agree while global kernels differ by boundary conditions or line-bundle holonomy. In systems with internal degrees of freedom, characters are replaced by unitary representations of path or gluing groupoids. In later applications of the same kernel-completion logic, analogous questions arise for gauge edge sectors and for temporal process-memory holonomy, where the forgotten data need not be a scalar phase.

This convergence is the main reason the boundary-kernel viewpoint is more than a reversal of Hamilton-Jacobi mechanics. Hamilton-Jacobi theory assumes that the action and relevant boundary variables have already been supplied. The boundary-completion question is different: when a proposed boundary description fails to compose, what additional boundary data must be present? In finite operational models the missing data may be a classical or quantum memory sector; in gauge systems an edge or centre algebra; in gravitational corner problems a codimension-two angle variable; in stochastic thermodynamics an affinity-resolving transition boundary; and in process tensor settings a memory system. The present paper proves the topological scalar instance of this pattern.

These consequences are structural rather than phenomenological. The scalar topological theorem shows that the Hamilton-Jacobi projection has nontrivial fibers; richer boundary sectors are natural extensions of the same compositional obstruction.

8 limits and relation to standard mechanics

The theorem concerns scalar topological sectors in closed amplitude kernels. In a simply connected configuration space, with one local branch, trivial holonomy, no internal sector, no caustic data, and no subleading amplitude information, the kernel-level description may reduce to ordinary Hamilton-Jacobi response geometry. In that regime the projection Π_{HJ} discards no topological character data because there are none.

Outside that regime, local Hamilton-Jacobi data are a quotient of the kernel rather than an equivalent description. The quotient can forget prefactors, Maslov indices, branch sums, flat holonomies, and other global or algebraic data. The scalar topological theorem establishes that the reverse implication from local Hamilton-Jacobi data to full boundary-kernel composition fails.

For the ring, the missing completion is a $U(1)$ character. In richer reversible amplitude settings, a natural analogue is a unitary representation of global gluing classes. Positive stochastic kernels or open irreversible systems can instead require non-unitary completion data.

9 conclusion

Composable boundary kernels project to local Hamilton-Jacobi data in regular semiclassical regimes, but the projection is not invertible. Multiply connected configuration spaces supply a concrete theorem: distinct character-labelled kernels K_χ have the same local Hamilton-Jacobi projection and different global composition data. The Aharonov-Bohm ring is the standard witness. Hamilton-Jacobi theory captures a local sharp shadow of kernel-level composition, not the full boundary-kernel object. Together with the sharp-limit reconstruction of response geometry from composable region weights, this establishes a precise sense in which boundary-kernel composition generalises Hamilton-Jacobi theory.

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